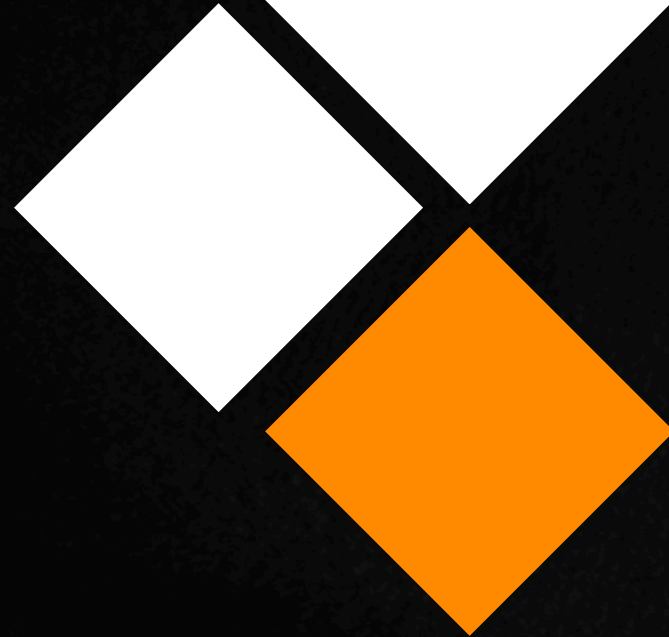


PIZZA

SALES

ANALYSIS



PROJECT OVERVIEW

Understanding sales performance is crucial for restaurants to optimize inventory, pricing, and promotional strategies. Sales analysis helps identify customer preferences, peak demand periods, and revenue-driving products to improve profitability and operational efficiency.

This project analyzes pizza sales data to identify revenue trends, top-selling products, customer ordering patterns, and seasonal demand variations. Using SQL for data analysis and Power BI for visualization, I built an interactive dashboard to uncover key performance metrics such as total revenue, average order value, peak sales hours, and best-performing pizza categories. The insights support data-driven decisions to optimize inventory, pricing strategies, and promotional campaigns.



PROBLEM STATEMENT

- The business is struggling to identify which pizza categories and products generate the highest revenue due to limited visibility into detailed sales performance.
- Operational decisions such as inventory planning and staffing are inefficient because peak sales hours and demand patterns are not clearly analyzed.
- Promotional strategies are underperforming as customer ordering behavior and seasonal trends are not effectively utilized to drive targeted marketing efforts.

DATA SET

The dataset contains pizza sales information such as :

- Order Details: Order ID, Order Date, Order Time, Quantity, Pizza ID
- Time Information: Date, Time (used to derive Day, Month, Hour, Peak Hours)
- Product Information: Pizza ID, Pizza Name, Category, Size, Price, Ingredients
- Sales Information: Quantity Sold, Unit Price, Total Revenue (derived)
- Relational Structure: Orders table linked with Order Details, Pizzas, and Pizza Types for complete transaction-level analysis

ANALYSIS

What is the Total Revenue?

```
select format(sum(price*quantity),2) as Total_revenue  
from order_details as o  
join pizzas as p  
on o.pizza_id = p.pizza_id;
```

Total_revenue

817,860.05

What is the Average Order Value?

```
select concat('$',format(sum(price*quantity)/count(distinct order_id),2)) as Average_order_value
from order_details as o
join pizzas as p
on o.pizza_id = p.pizza_id;
```

Average_order_value

\$38.31

How many Pizza's were sold?

```
select format(sum(quantity),0) as Total_quantity  
from order_details;
```

Total_quantity
49,574

How many orders were placed?

```
select format(count(distinct order_id),0) as Total_orders  
from order_details;
```

Total_orders
21,350

What is the Average Pizza's Per Order?

```
select round(sum(quantity)/count(distinct order_id)) as Average_per_order  
from order_details;
```

Average_per_order
2

Which day of the week we got most orders throughout the year?

```
select dayname(`date`) as Week_day,format(count(distinct order_id),0) as Total_orders
from orders
group by Week_day
order by Total_orders desc;
```

Week_day	Total_orders
Friday	3,538
Thursday	3,239
Saturday	3,158
Wednesday	3,024
Tuesday	2,973
Monday	2,794
Sunday	2,624

Which month we got most orders?

```
select monthname(`date`) as Month_name,format(count(distinct order_id),0) as Total_orders  
from orders  
group by Month_name  
order by Total_orders desc;
```

Month_name	Total_orders
July	1,935
May	1,853
January	1,845
August	1,841
March	1,840
April	1,799
November	1,792
June	1,773
February	1,685
December	1,680
September	1,661
October	1,646

What percentage of sales contributed by each Pizza Category?

```
with Overall_sales as
(select format(sum(price*quantity),2) as Total_sales
 from order_details as o
  join pizzas as p
   on o.pizza_id = p.pizza_id),

sales_by_category as
(select t.category,
      format(sum(price*quantity),2) as category_sales
 from order_details as o
  join pizzas as p
   on o.pizza_id = p.pizza_id
  join pizza_types as t
   on p.pizza_type_id = t.pizza_type_id
 group by t.category)

select Category,
       concat('$',category_sales) as Total_sales,
       concat(round(100*(category_sales/Total_sales),2),'%') as Percentage_contribution
 from Overall_sales,sales_by_category
 order by percentage_contribution desc;
```

Category	Total_sales	Percentage_contribution
Classic	\$220,053.10	26.93%
Supreme	\$208,197.00	25.46%
Chicken	\$195,919.50	23.87%
Veggie	\$193,690.45	23.62%

How many Pizza's were sold by each Category?

```
select Category,format(sum(quantity),0) as Total_quantity
from order_details as o
  join pizzas as p
    on o.pizza_id = p.pizza_id
  join pizza_types as t
    on p.pizza_type_id = t.pizza_type_id
group by category
order by Total_quantity desc;
```

Category	Total_quantity
Classic	14,888
Supreme	11,987
Veggie	11,649
Chicken	11,050

What are the Top 5 Pizza's by revenue?

```
select `name` as Name_of_Pizza,concat('$',format(sum(price*quantity),0)) as Total_revenue
from order_details as o
join pizzas as p
on o.pizza_id = p.pizza_id
join pizza_types as t
on p.pizza_type_id = t.pizza_type_id
group by Name_of_Pizza
order by Total_revenue desc
limit 5;
```

Name_of_Pizza	Total_revenue
The Thai Chicken Pizza	\$43,434
The Barbecue Chicken Pizza	\$42,768
The California Chicken Pizza	\$41,410
The Classic Deluxe Pizza	\$38,180
The Spicy Italian Pizza	\$34,831

What are the Bottom 5 Pizza's by revenue?

```
select `name` as Name_of_Pizza,concat('$',format(sum(price*quantity),0)) as Total_revenue
from order_details as o
join pizzas as p
  on o.pizza_id = p.pizza_id
join pizza_types as t
  on p.pizza_type_id = t.pizza_type_id
group by Name_of_Pizza
order by Total_revenue asc
limit 5;
```

Name_of_Pizza	Total_revenue
The Brie Carre Pizza	\$11,588
The Green Garden Pizza	\$13,956
The Spinach Supreme Pizza	\$15,278
The Mediterranean Pizza	\$15,360
The Spinach Pesto Pizza	\$15,596

What are the Top 5 Pizza's by Orders?

```
select `name` as Name_of_Pizza, count(distinct order_id) as Total_orders
from order_details as o
join pizzas as p
  on o.pizza_id = p.pizza_id
join pizza_types as t
  on p.pizza_type_id = t.pizza_type_id
group by Name_of_Pizza
order by Total_orders desc
limit 5;
```

Name_of_Pizza	Total_orders
The Classic Deluxe Pizza	2329
The Hawaiian Pizza	2280
The Pepperoni Pizza	2278
The Barbecue Chicken Pizza	2273
The Thai Chicken Pizza	2225

What are the Bottom 5 Pizza's by Orders?

```
with cte as
(select `name` as Name_of_Pizza,count(distinct order_id) as Total_orders,
dense_rank() over(order by count(distinct order_id) asc) as Ranks
from order_details as o
join pizzas as p
on o.pizza_id = p.pizza_id
join pizza_types as t
on p.pizza_type_id = t.pizza_type_id
group by `name`)

select *
from cte
where Ranks <= 5;
```

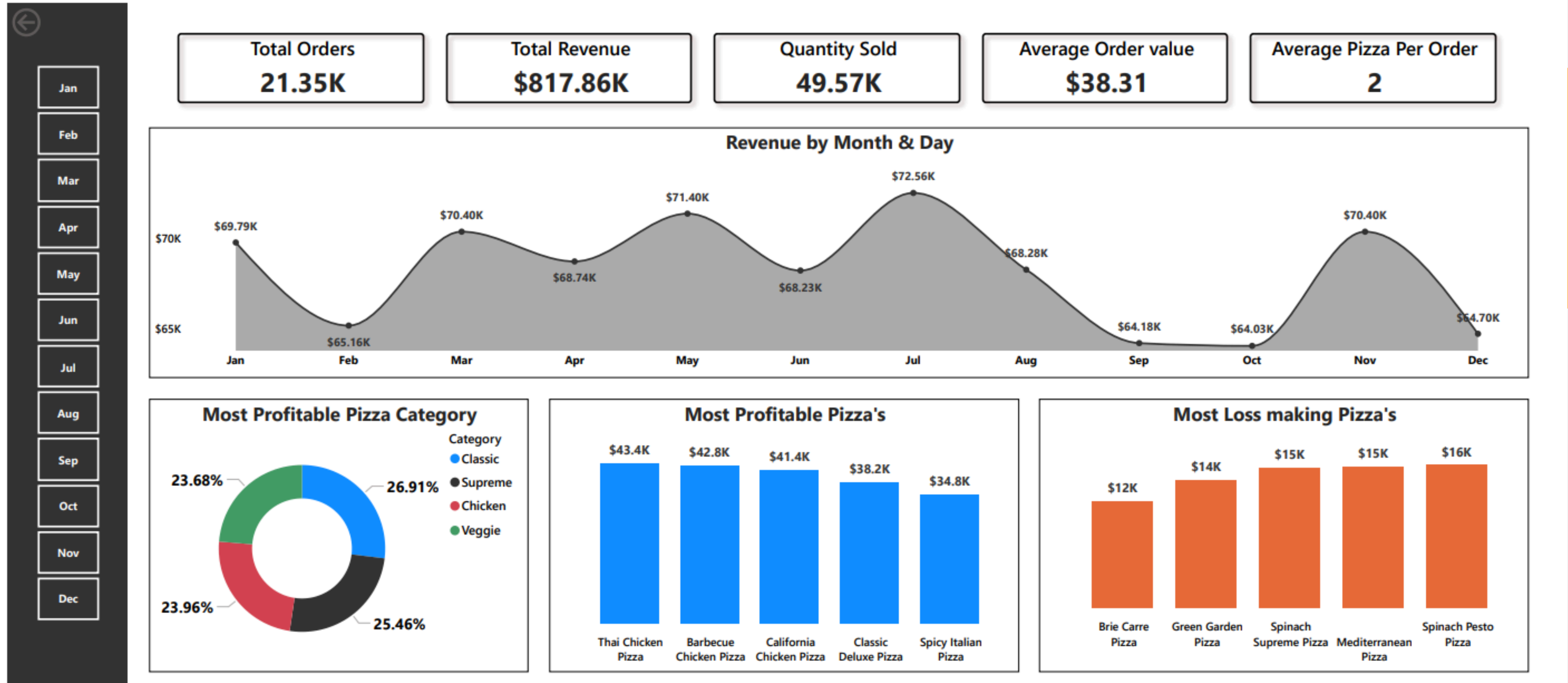
Name_of_Pizza	Total_orders	Ranks
The Brie Carre Pizza	480	1
The Mediterranean Pizza	912	2
The Calabrese Pizza	918	3
The Spinach Supreme Pizza	918	3
The Chicken Pesto Pizza	938	4
The Soppressata Pizza	941	5

Which Time most Pizza's were sold?

```
select
case when time_format(`time`, '%H') between 09 and 11 then 'Morning'
      when time_format(`time`, '%H') between 12 and 17 then 'Afternoon'
      when time_format(`time`, '%H') between 18 and 23 then 'Evening'
      else 'Night' end as Time_of_day,
      format(sum(quantity),0) as Total_pizzas
from order_details as d
join orders as o
  on o.order_id = d.order_id
group by Time_of_day
order by Time_of_day;
```

Time_of_day	Total_pizzas
Afternoon	29,468
Evening	17,356
Morning	2,750

DASHBOARDS



Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

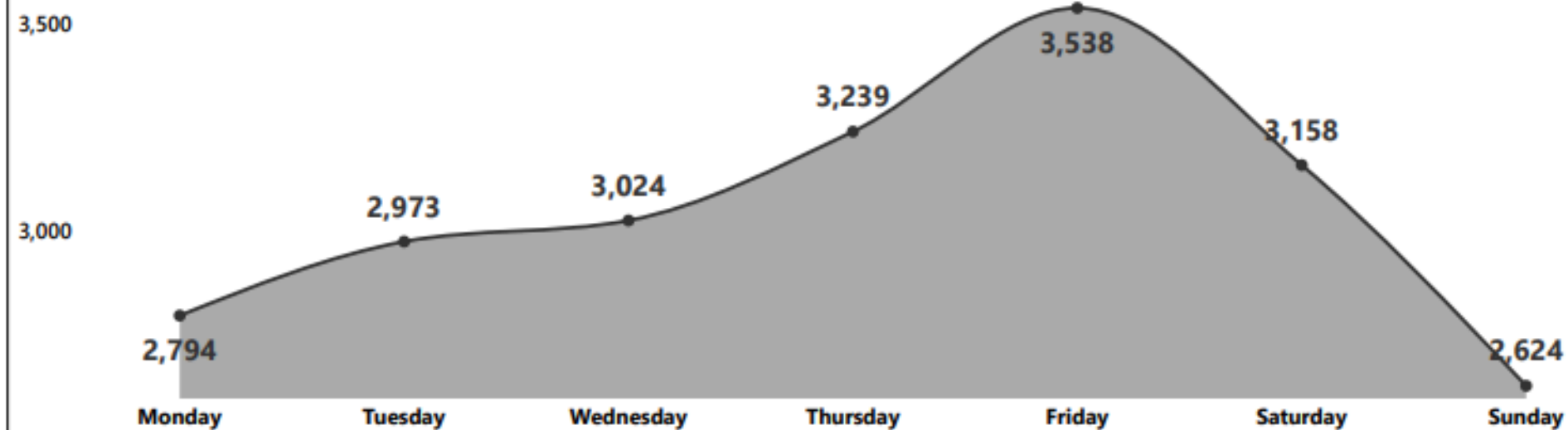
Sep

Oct

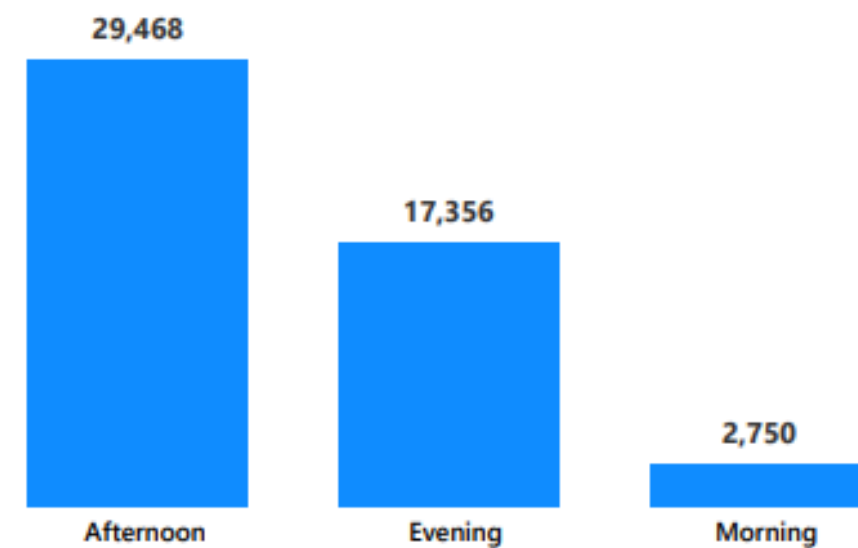
Nov

Dec

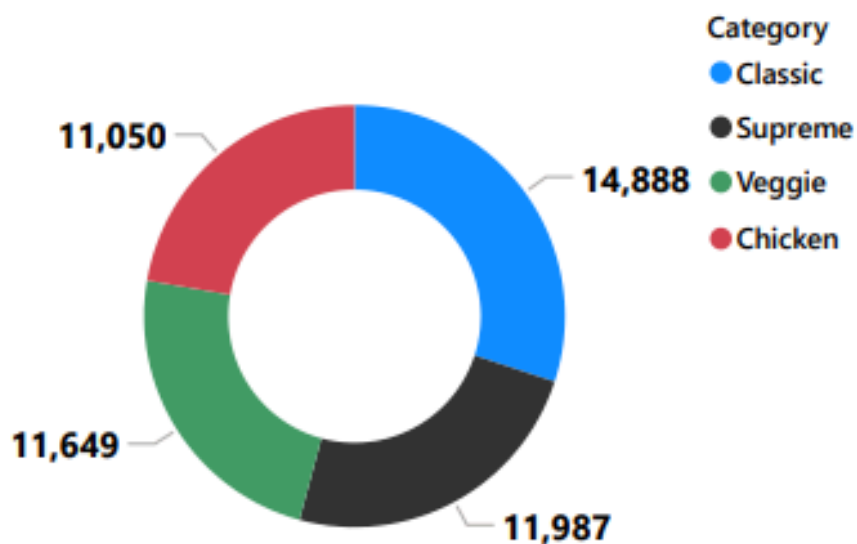
Orders by Month & Day



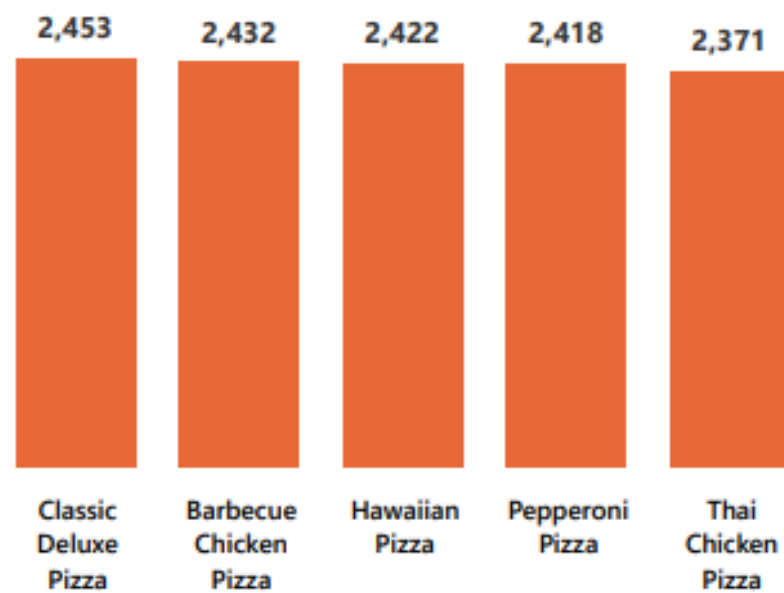
Quantity Sold by Time of day



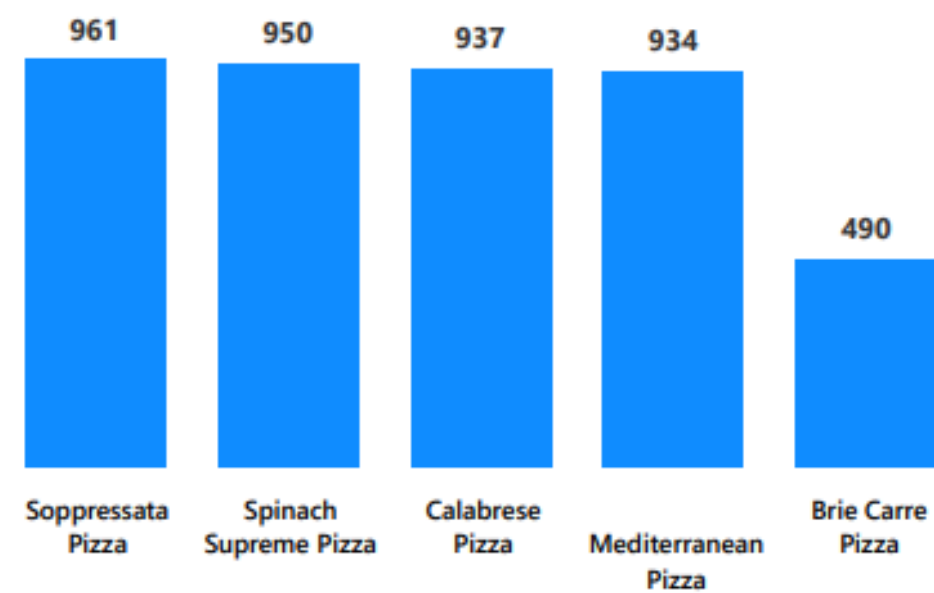
Quantity Sold Pizza Category



Most Sold Pizza's



Least Sold Pizza's



BUSINESS INSIGHTS

- 
- Revenue distribution varies across pizza categories, with the Classic category contributing the highest share, indicating strong and consistent customer preference.
 - Sales peak during afternoon hours and weekends, highlighting critical time periods for staffing optimization and inventory planning.
 - A small group of top-selling pizzas drives a significant portion of total revenue, creating dependency on key menu items.
 - Several low-performing pizzas generate minimal revenue, suggesting opportunities for menu optimization or targeted promotions.
 - Monthly revenue remains relatively stable throughout the year, indicating steady demand with minor seasonal fluctuations that can be leveraged through strategic campaigns.

SUGGESTIONS

BOOST BEST- SELLER REVENUE

Prioritize and promote top-selling pizzas through combo deals and upselling strategies to maximize revenue per order.

OPTIMIZE PEAK OPERATIONS

Optimize staffing and inventory during afternoon and weekend peak hours to improve operational efficiency and customer experience.

IMPROVE MENU PROFITABILITY

Reassess low-performing pizzas by adjusting pricing, offering promotions, or removing underperforming items to enhance overall profitability.



ESTIMATED IMPACT



- **INCREASE AVERAGE ORDER VALUE**
Promoting top-selling pizzas through combos and upselling can increase average order value by 8–15% and overall revenue by 5–10%.
- **IMPROVE OPERATIONAL EFFICIENCY**
Optimizing staffing and inventory during peak hours can improve operational efficiency by 10–18% and reduce service delays by 15–20%.
- **ENHANCE PROFIT MARGINS**
Adjusting pricing or removing low-performing pizzas can improve overall profit margins by 6–12%.
- **BOOST OFF - PEAK SALES**
Targeted promotions during slow hours can increase off-peak sales by 7–14%, stabilizing daily revenue flow.



THANK
YOU

- SHRISHAIL PACHUDE